MH1101 Tutorial 12 (Week 13)

Solution

Reference: Chapter 6.3, 6.4, 6.5 (Lecture Notes)

1. Find the Taylor series for $f(x)$ centred at the given value of a. Assume that f has a power series expansion. (Do not show that $R_n(x) \to 0$). Also find the associated radius of convergence.

(a) $f(x) = \ln x$, $a = 2$

(b) $f(x) = e^{2x}$, $a = 3$

(c) $f(x) = \cos x$, $a = \pi/2$

(d) $f(x) = \sqrt{x}$, $a = 16$

(e) $f(x) = \dfrac{1}{7x - 13}$, $a = 2$.

Solution

(a) We have

n	$f^{(n)}(x)$	$f^{(n)}(2)$
0	$\ln x$	$\ln 2$
1	$1/x$	$1/2$
2	$-1/x^2$	$-1/2^2$
3	$2/x^3$	$2/2^3$
4	$-6/x^4$	$-6/2^4$
5	$24/x^5$	$24/2^5$
$\vdots$	$\vdots$	$\vdots$

$$
\begin{aligned}
f(x) = \ln x &= \sum_{n=0}^{\infty} \frac{f^{(n)}(2)}{n!}(x-2)^n \\
&= \frac{\ln 2}{0!}(x-2)^0 + \frac{1/2}{1!}(x-2)^1 + \frac{-1/2^2}{2!}(x-2)^2 \\
&\quad + \frac{2/2^3}{3!}(x-2)^3 + \frac{-6/2^4}{4!}(x-2)^4 + \frac{24/2^5}{5!}(x-2)^5 + \cdots \\
&= \ln 2 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(n-1)!}{n!2^n}(x-2)^n \\
&= \ln 2 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n2^n}(x-2)^n
\end{aligned}
$$

$$
\begin{aligned}
\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n\to\infty} \left| \frac{(-1)^{n+2}(x-2)^{n+1}}{(n+1)2^{n+1}} \cdot \frac{n2^n}{(-1)^{n+1}(x-2)^n} \right| \\
&= \lim_{n\to\infty} \left| \frac{(-1)(x-2)n}{(n+1)2} \right| \\
&= \lim_{n\to\infty} \left(\frac{n}{n+1} \right) \frac{|x-2|}{2} \\
&= \frac{|x-2|}{2}
\end{aligned}
$$

By the Ratio Test, the series converges for $\frac{|x-2|}{2} < 1 \iff |x-2| < 2$ and diverges if $|x-2| > 2$, i.e. $R = 2$.

(b) We have

n	$f^{(n)}(x)$	$f^{(n)}(3)$
0	e^{2x}	e^6
1	$2e^{2x}$	$2e^6$
2	2^2e^{2x}	$4e^6$
3	2^3e^{2x}	$8e^6$
4	2^4e^{2x}	$16e^6$
$\vdots$	$\vdots$	$\vdots$

$$
\begin{aligned}
f(x) = e^{2x} &= \sum_{n=0}^{\infty} \frac{f^{(n)}(3)}{n!}(x-3)^n \\
&= \frac{e^6}{0!}(x-3)^0 + \frac{2e^6}{1!}(x-3)^1 + \frac{4e^6}{2!}(x-3)^2 \\
&\quad + \frac{8e^6}{3!}(x-3)^3 + \frac{16e^6}{4!}(x-3)^4 + \cdots \\
&= \sum_{n=0}^{\infty} \frac{2^n e^6}{n!}(x-3)^n.
\end{aligned}
$$

$$
\begin{aligned}
\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n\to\infty} \left| \frac{2^{n+1}e^6(x-3)^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n e^6 (x-3)^n} \right| \\
&= \lim_{n\to\infty} \frac{2}{n+1}|x-3| = 0 < 1, \quad \text{for all } x.
\end{aligned}
$$

So $R = \infty$.

(c) We have

n	$f^{(n)}(x)$	$f^{(n)}(\pi/2)$
0	$\cos x$	0
1	$-\sin x$	-1
2	$-\cos x$	0
3	$\sin x$	1
4	$\cos x$	0
5	$-\sin x$	-1
6	$-\cos x$	0
7	$\sin x$	1
$\vdots$	$\vdots$	$\vdots$

$$
\begin{aligned}
f(x) = \cos x &= \sum_{n=0}^{\infty} \frac{f^{(n)}(\pi/2)}{n!}\left(x - \frac{\pi}{2}\right) \\
&= \frac{-1}{1!}\left(x - \frac{\pi}{2}\right)^1 + \frac{1}{3!}\left(x - \frac{\pi}{2}\right)^3 + \frac{-1}{5!}\left(x - \frac{\pi}{2}\right)^5 \\
&\quad + \frac{1}{7!}\left(x - \frac{\pi}{2}\right)^7 + \cdots \\
&= \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(2n+1)!}\left(x - \frac{\pi}{2}\right)^{2n+1}
\end{aligned}
$$

$$
\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(-1)^{n+2}}{(2n+3)!}\left(x - \frac{\pi}{2}\right)^{2n+3} \cdot \frac{(2n+1)!}{(-1)^{n+1}\left(x - \frac{\pi}{2}\right)^{2n+1}}\right| \\
&= \lim_{n\to\infty} \frac{\left(x - \frac{\pi}{2}\right)^2}{(2n+3)(2n+2)} \\
&= 0 < 1 \quad \text{for all } x.
\end{aligned}
$$

So $R = \infty$.

(d) We have

n	$f^{(n)}(x)$	$f^{(n)}(16)$
0	$\sqrt{x}$	4
1	$\frac{1}{2}x^{-1/2}$	$\frac{1}{2} \cdot \frac{1}{4}$
2	$-\frac{1}{4}x^{-3/2}$	$-\frac{1}{4} \cdot \frac{1}{4^3}$
3	$\frac{3}{8}x^{-5/2}$	$\frac{3}{8} \cdot \frac{1}{4^5}$
4	$-\frac{15}{16}x^{-7/2}$	$-\frac{15}{16} \cdot \frac{1}{4^7}$
$\vdots$	$\vdots$	$\vdots$

$$
\begin{aligned}
f(x)=\sqrt{x} &= \sum_{n=0}^{\infty}\frac{f^{(n)}(16)}{n!}(x-16)^n \\
&= \frac{4}{0!}(x-16)^0+\frac{1}{2}\cdot\frac{1}{4}\cdot\frac{1}{1!}(x-16)^1-\frac{1}{4}\cdot\frac{1}{4^3}\cdot\frac{1}{2!}(x-16)^2 \\
&\quad +\frac{3}{8}\cdot\frac{1}{4^5}\cdot\frac{1}{3!}(x-16)^3-\frac{15}{16}\cdot\frac{1}{4^7}\cdot\frac{1}{4!}(x-16)^4+\cdots \\
&= 4+\frac{1}{8}(x-16)+\sum_{n=2}^{\infty}(-1)^{n-1}\frac{1\cdot 3\cdot 5\cdots(2n-3)}{2^n4^{2n-1}n!}(x-16)^n \\
&= 4+\frac{1}{8}(x-16)+\sum_{n=2}^{\infty}(-1)^{n-1}\frac{1\cdot 3\cdot 5\cdots(2n-3)}{2^{5n-2}n!}(x-16)^n
\end{aligned}
$$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|$$

$$
\begin{aligned}
&= \lim_{n\to\infty}\left|(-1)^n\frac{1\cdot 3\cdot 5\cdots(2n-1)(x-16)^{n+1}}{2^{5n+3}(n+1)!}\cdot\frac{2^{5n-2}n!}{(-1)^{n-1}1\cdot 3\cdot 5\cdots(2n-3)(x-16)^n}\right| \\
&= \lim_{n\to\infty}\frac{(2n-1)|x-16|}{2^5(n+1)} \\
&= \lim_{n\to\infty}\left(\frac{2-\frac{1}{n}}{1+\frac{1}{n}}\right)\frac{|x-16|}{32} \\
&= 2\cdot\frac{|x-16|}{32} \\
&= \frac{|x-16|}{16}.
\end{aligned}
$$

By the Ratio Test, the series converges if $\frac{|x-16|}{16}<1 \iff |x-16|<16$ and diverges if $|x-16|>16$, so $R=16$.

(e) Instead of finding the Taylor series using the definition, we can manipulate a geometric series.

$$\begin{aligned}
\frac{1}{7x+2} &= \frac{1}{7(x-2)+14-13} \\
&= \frac{1}{1+7(x-2)} \\
&= \frac{1}{1-(-7(x-2))} \\
&= \sum_{n=0}^{\infty}(-1)^n 7^n (x-2)^n.
\end{aligned}$$

Since we manipulated a geometric series, this series converges for $|7(x-2)| < 1 \iff |x-2| < \frac{1}{7} \iff \frac{13}{7} < x < \frac{15}{7}$. So $R = \frac{1}{7}$. This means that the function $\frac{1}{7x+2}$ has a power series representation in $x-2$ (centred at 2) on $\frac{13}{7} < x < \frac{15}{7}$. By the uniqueness of power series expansion, this series is also the Taylor series for the function at $x = 2$ (see Theorem 3).

□

2. Prove that the series obtained in Q1(c) represents $\cos x$ for all x.

Solution Let $f(x) = \cos x$. Then $f^{(n)}(x) = \pm\cos x$ or $\pm\sin x$. So $\left|f^{(n)}(x)\right| \leq 1$ for all x. Apply Error Bound with $M = 1$, we have

$$0 \leq |R_n(x)| \leq \frac{M}{(n+1)!}\left|x-\frac{\pi}{2}\right|^{n+1} = \frac{1}{(n+1)!}\left|x-\frac{\pi}{2}\right|^{n+1}.$$

Since $\frac{1}{(n+1)!}\left|x-\frac{\pi}{2}\right|^{n+1} \to 0$ as $n \to \infty$, we deduce by Squeeze Theorem that $|R_n(x)| \to 0$ as $n \to 0$. That is, the series represents $\cos x$ for all x. □

3. Find the first three nonzero terms in the Maclaurin series for the function.

(a) $f(x) = \sec x$

(b) $f(x) = e^x \ln(1+x)$.

Solution

(a) Note that

$$\sec x = \frac{1}{\cos x} = \frac{1}{1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \cdots}.$$

By Long division, we have

$$\begin{array}{r|l}
 & 1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \cdots \\ \cline{2-2}
1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \cdots & 1 \\
 & 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \cdots \\ \cline{2-2}
 & \quad \frac{1}{2}x^2 - \frac{1}{24}x^4 + \cdots \\
 & \quad \frac{1}{2}x^2 - \frac{1}{4}x^4 + \cdots \\ \cline{2-2}
 & \qquad \frac{5}{24}x^4 + \cdots \\
 & \qquad \frac{5}{24}x^4 + \cdots \\ \cline{2-2}
 & \qquad\qquad \cdots
\end{array}$$

Hence,

$$\sec x = 1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \cdots .$$

(b) Recall that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

Therefore,

$$e^x \ln(1+x) \;=\; \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots\right)$$

Writing only terms with degree ≤ 3, we have

$$e^x \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + x^2 - \frac{x^3}{2} + \frac{x^3}{2} + \cdots = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots$$

$\square$

4. Use series to evaluate the limit.

(a) $\displaystyle\lim_{x\to 0}\frac{2x-\ln(1+2x)}{x^2}$

(b) $\displaystyle\lim_{x\to 0}\frac{\sqrt{1+x}-1-\frac{1}{2}x}{x^2}$

Solution

(a)

$$\begin{aligned}
\lim_{x\to 0}\frac{2x-\ln(1+2x)}{x^2} &= \lim_{x\to 0}\frac{2x-\left(2x-\frac{(2x)^2}{2}+\frac{(2x)^3}{3}-\frac{(2x)^4}{4}+\cdots\right)}{x^2}\\
&= \lim_{x\to \infty}\frac{\frac{(2x)^2}{2}-\frac{(2x)^3}{3}+\frac{(2x)^4}{4}-\cdots}{x^2}\\
&= \lim_{x\to 0}\left(\frac{4}{2}-\frac{8x}{3}+\frac{16x^2}{4}-\cdots\right)\\
&= \frac{4}{2}=2.
\end{aligned}$$

(b) Using Binomial series,

$$\sqrt{1+x}=1+\frac{1}{2}x+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}x^2+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!}x^3+\cdots$$

$$\begin{aligned}
\lim_{x\to 0}\frac{\sqrt{1+x}-1-\frac{1}{2}x}{x^2} &= \lim_{x\to 0}\frac{\left(1+\frac{1}{2}x-\frac{1}{8}x^2+\frac{1}{16}x^3-\cdots\right)-1-\frac{1}{2}x}{x^2}\\
&= \lim_{x\to 0}\frac{\left(-\frac{1}{8}x^2+\frac{1}{16}x^3-\cdots\right)}{x^2}\\
&= \lim_{x\to 0}\left(-\frac{1}{8}+\frac{1}{16}x-\cdots\right)\\
&= -\frac{1}{8}.
\end{aligned}$$

$\square$

5. Use the binomial series to expand the function as a power series. State the radius of convergence.

$$f(x) = \sqrt[3]{8+x}.$$

Solution

$$\begin{aligned}
\sqrt[3]{8+x} &= \sqrt[3]{8\left(1+\frac{x}{8}\right)} \\
&= 2\left(1+\frac{x}{8}\right)^{1/3} \\
&= 2\sum_{n=0}^{\infty}\binom{1/3}{n}\left(\frac{x}{8}\right)^n \\
&= 2\left(1+\frac{1}{3}\left(\frac{x}{8}\right)+\frac{\frac{1}{3}\left(-\frac{2}{3}\right)}{2!}\left(\frac{x}{8}\right)^2+\frac{\frac{1}{3}\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3!}\left(\frac{x}{8}\right)^3+\cdots\right) \\
&= 2\left(1+\frac{x}{24}+\sum_{n=2}^{\infty}\frac{(-1)^{n-1}(2\cdot 5\cdots(3n-4))}{3^n 8^n n!}x^n\right)
\end{aligned}$$

The series converges if $\left|\frac{x}{8}\right| < 1$ and diverges if $\left|\frac{x}{8}\right| > 1$. So $R = 8$. $\square$

6. Find the sum of the series.

(a) $\displaystyle\sum_{n=0}^{\infty}\frac{(-1)^n\pi^{2n+1}}{4^{2n+1}(2n+1)!}$

(b) $\displaystyle\frac{1}{1\cdot 2}-\frac{1}{3\cdot 2^3}+\frac{1}{5\cdot 2^5}-\frac{1}{7\cdot 2^7}+\cdots$

Solution

(a)

$$\sum_{n=0}^{\infty}\frac{(-1)^n\pi^{2n+1}}{4^{2n+1}(2n+1)!} = \sum_{n=0}^{\infty}\frac{(-1)^n\left(\frac{\pi}{4}\right)^{2n+1}}{(2n+1)!} = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}.$$

(b)

$$\begin{aligned}
&\frac{1}{1\cdot 2} - \frac{1}{3\cdot 2^3} + \frac{1}{5\cdot 2^5} - \frac{1}{7\cdot 2^7} + \cdots \\
&= \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)2^{2n+1}} \\
&= \sum_{n=0}^{\infty} (-1)^n \frac{(1/2)^{2n+1}}{(2n+1)} \\
&= \tan^{-1}\left(\frac{1}{2}\right).
\end{aligned}$$

□

7. Show that if p is an n-th degree polynomial, then

$$p(x+1) = \sum_{i=0}^{n} \frac{p^{(i)}(x)}{i!}.$$

Solution Since the degree of the polynomial is n, we have

$$p^{(i)}(x) = 0, \quad \text{for all} \quad i > n.$$

So its Taylor series at a is

$$p(x) = \sum_{i=0}^{n} \frac{p^{(i)}(a)}{i!}(x-a)^i.$$

Substituting $x = a + 1$, we have

$$p(a+1) = \sum_{i=0}^{n} \frac{p^{(i)}(a)}{i!}.$$

But a is arbitrary, the above is true for all a, so replacing a by the variable x, we have

$$p(x+1) = \sum_{i=0}^{n} \frac{p^{(i)}(x)}{i!}.$$

□

8.

(a) Show that the Maclaurin series of the function

$$f(x) = \frac{x}{1 - x - x^2} \quad \text{is} \quad \sum_{n=1}^{\infty} f_n x^n$$

where f_n is the nth **Fibonacci number**, that is, $f_1 = 1$, $f_2 = 1$, and

$$f_n = f_{n-1} + f_{n-2} \quad \text{for all} \quad n \geq 3.$$

(b) By writing $f(x)$ as a sum of partial fraction and thereby obtaining the Maclaurin series in a different way, find an explicit formula for the nth Fibonacci number.

Solution

(a) We will show in part (b) (by manipulating geometric series) that $f(x)$ is equal to its Maclaurin series for x on some interval. For the time being, let us assume that this is true:

$$f(x) = \frac{x}{1 - x - x^2} = c_0 + c_1 x + c_2 x^2 + \cdots .$$

where $c_0 + c_1 x + c_2 x^2 + \cdots$ is the Maclaurin series for $f(x)$. Then

$$\begin{aligned} x &= (1 - x - x^2)\left(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots\right) \\ &= c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots \\ &\qquad - c_0 x - c_1 x^2 - c_2 x^3 - c_3 x^4 - \cdots \\ &\qquad\qquad - c_0 x^2 - c_1 x^3 - c_2 x^4 - \cdots \\ x &= c_0 + (c_1 - c_0)x + (c_2 - c_1 - c_0)x^2 + (c_3 - c_2 - c_1)x^3 + \cdots \end{aligned}$$

Comparing coefficients on both sides, we must have

$$\begin{aligned} c_0 &= 0 \\ c_1 - c_0 &= 1 \\ c_2 - c_1 - c_0 &= 0 \\ c_3 - c_2 - c_1 &= 0 \\ &\vdots \\ c_n - c_{n-1} - c_{n-2} &= 0 \end{aligned}$$

This implies that $c_0 = 0$, $c_1 = 1$, $c_2 = 1$, $c_3 = c_2 + c_1 = 2$, and for all $n \geq 3$,

$$c_n = c_{n-1} + c_{n-2}.$$

Each c_n is equal to the nth Fibonacci number by definition, that is $c_n = f_n$ for all $n \geq 1$. Since $c_0 = 0$, we have

$$f(x) = \sum_{n=0}^{\infty} c_n x^n = \sum_{n=1}^{\infty} c_n x^n = \sum_{n=1}^{\infty} f_n x^n.$$

(b) Completing the squares on $x^2 + x - 1$:

$$\begin{aligned} x^2 + x - 1 &= \left(x + \frac{1}{2}\right)^2 - \frac{5}{4} \\ &= \left(x + \frac{1}{2}\right)^2 - \left(\frac{\sqrt{5}}{2}\right)^2 \\ &= \left(x + \frac{1}{2} - \frac{\sqrt{5}}{2}\right)\left(x + \frac{1}{2} + \frac{\sqrt{5}}{2}\right) \\ &= (x+\alpha)(x+\beta), \end{aligned}$$

where

$$\alpha = \frac{1-\sqrt{5}}{2}, \quad \beta = \frac{1+\sqrt{5}}{2}.$$

Therefore,

$$\begin{aligned} \frac{x}{1 - x - x^2} &= \frac{-x}{x^2 + x - 1} \\ &= \frac{-x}{(x+\alpha)(x+\beta)}. \end{aligned}$$

The partial fraction decomposition of $\dfrac{-x}{(x+\alpha)(x+\beta)}$ is

$$\frac{-x}{(x+\alpha)(x+\beta)} = \frac{A}{x+\alpha} + \frac{B}{x+\beta}.$$

$$-x = A(x+\beta) + B(x+\alpha) = (A+B)x + A\beta + B\alpha.$$

This implies that

$$A + B = -1, \quad A\beta + B\alpha = 0$$
$$\implies \quad A = \frac{\alpha}{\beta - \alpha}, \quad B = \frac{-\beta}{\beta - \alpha}.$$

$$\begin{aligned}
\frac{A}{x+\alpha} &= \frac{\frac{\alpha}{\beta-\alpha}}{x+\alpha} \\
&= \frac{1}{\beta-\alpha} \cdot \frac{1}{1-(-x/\alpha)} \\
&= \frac{1}{\beta-\alpha} \sum_{n=0}^{\infty} \left(-\frac{x}{\alpha}\right)^n, \quad \forall |x| < |\alpha|
\end{aligned}$$

Likewise,

$$\begin{aligned}
\frac{B}{x+\beta} &= \frac{\frac{-\beta}{\beta-\alpha}}{x+\beta} \\
&= -\frac{1}{\beta-\alpha} \cdot \frac{1}{1-(-x/\beta)} \\
&= -\frac{1}{\beta-\alpha} \cdot \sum_{n=0}^{\infty} \left(-\frac{x}{\beta}\right)^n, \quad \forall |x| < |\beta|
\end{aligned}$$

Hence, on the values of x for which both geometric series above converge (e.g. we can take $|x| < \min\{|\alpha|, |\beta|\}$), we have

$$\begin{aligned}
\frac{x}{1-x-x^2} &= \frac{1}{\beta-\alpha} \sum_{n=0}^{\infty} \left(-\frac{x}{\alpha}\right)^n - \frac{1}{\beta-\alpha} \cdot \sum_{n=0}^{\infty} \left(-\frac{x}{\beta}\right)^n \\
&= \frac{1}{\beta-\alpha} \sum_{n=0}^{\infty} \left(\left(-\frac{x}{\alpha}\right)^n - \left(-\frac{x}{\beta}\right)^n\right) \\
&= \sum_{n=0}^{\infty} \frac{1}{\beta-\alpha} \cdot \left(\left(-\frac{1}{\alpha}\right)^n - \left(-\frac{1}{\beta}\right)^n\right) x^n
\end{aligned}$$

where the series on the right-hand-side must be the Maclaurin series for the function (by the Uniqueness of Power Series Representation of the function).

We deduce from part (a) that the nth Fibonacci number is

$$\begin{aligned} f_n &= \frac{1}{\beta-\alpha}\cdot\left(\left(-\frac{1}{\alpha}\right)^n-\left(-\frac{1}{\beta}\right)^n\right) \\ &= \frac{1}{\sqrt{5}}\cdot\left(\left(-\frac{2}{1-\sqrt{5}}\right)^n-\left(-\frac{2}{1+\sqrt{5}}\right)^n\right) \end{aligned}$$

$\square$

Answer Keys. **1.** (a) $\ln 2+\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n2^n}(x-2)^n,\ R=2$ (b) $\sum_{n=0}^{\infty}\frac{2^n e^6}{n!}(x-3)^n,\ R=\infty$ (c) $\sum_{n=0}^{\infty}\frac{(-1)^{n+1}}{(2n+1)!}\left(x-\frac{\pi}{2}\right)^{2n+1},\ R=\infty$ (d) $4+\frac{1}{8}(x-16)+\sum_{n=2}^{\infty}(-1)^{n-1}\frac{1\cdot3\cdot5\cdots(2n-3)}{2^{5n-2}n!}(x-16)^n$, $R=16$ (e) $\sum_{n=0}^{\infty}7^n(x-2)^n$, $R=\frac{1}{7}$.

3. (a) $\sec x=1+\frac{1}{2}x+\frac{5}{24}x^4+\cdots$. (b) $e^x\ln(1+x)=x+\frac{x^2}{2}+\frac{x^3}{3}+\cdots$ **4.** (a) 2 (b) $-\frac{1}{8}$ **6.** (a) $\frac{1}{\sqrt{2}}$ (b) $\tan^{-1}\left(\frac{1}{2}\right)$

8. (b) $f_n=\frac{1}{\sqrt{5}}\cdot\left(\left(-\frac{2}{1-\sqrt{5}}\right)^n-\left(-\frac{2}{1+\sqrt{5}}\right)^n\right)$